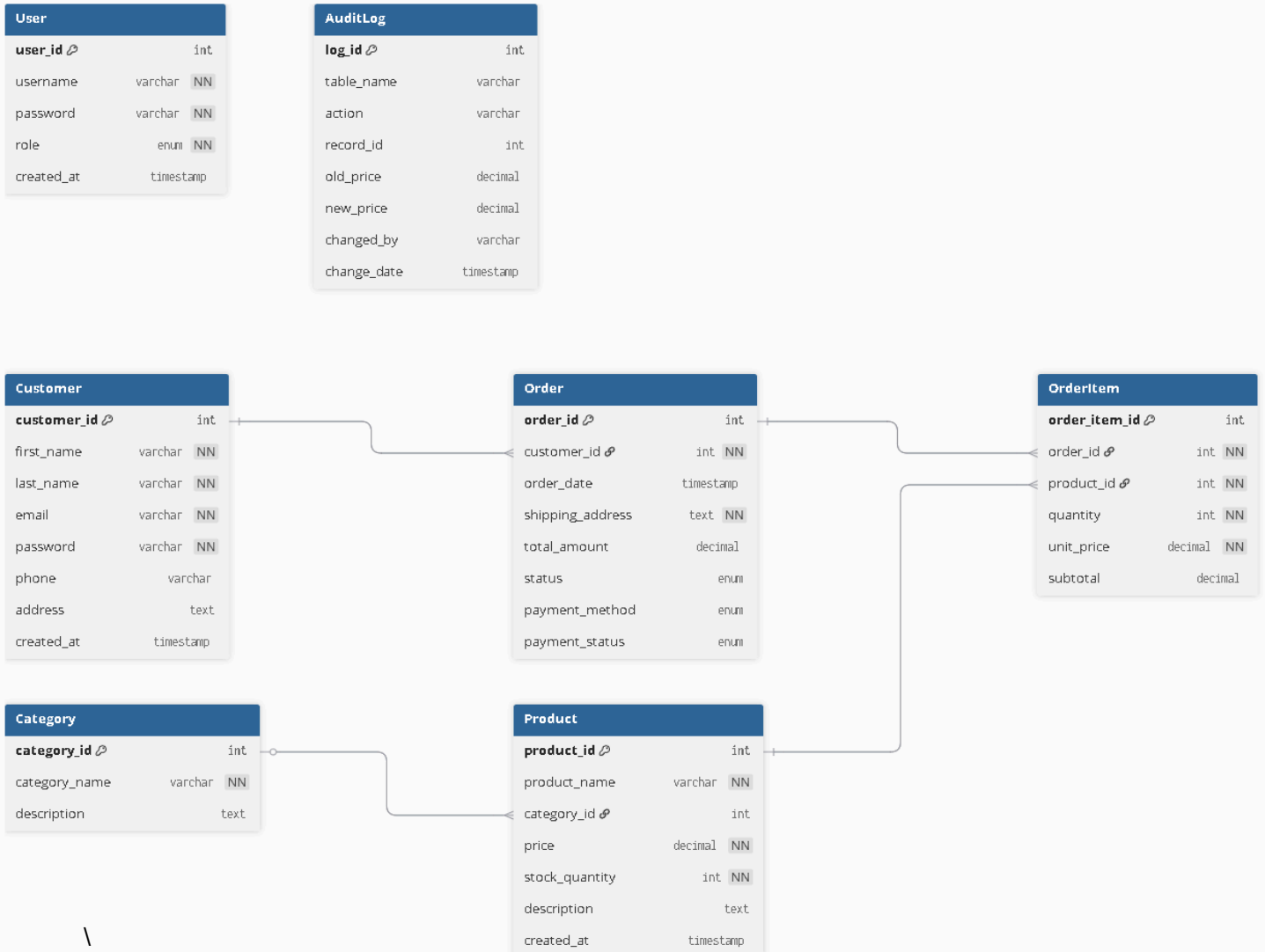




Final Documentation Compilation

Final ER Diagram



Database schema and table descriptions

```
-- =====  
-- ONLINE SHOP SYSTEM  
-- IT 105 Final Project  
-- =====
```

```
CREATE DATABASE OnlineShop;  
USE OnlineShop;
```

```
-- Table 1: Category
```

```
CREATE TABLE Category (  
    category_id INT PRIMARY KEY AUTO_INCREMENT,  
    category_name VARCHAR(50) UNIQUE NOT NULL,  
    description TEXT  
);
```

```
-- Table 2: Customer
```

```
CREATE TABLE Customer (  
    customer_id INT PRIMARY KEY AUTO_INCREMENT,  
    first_name VARCHAR(50) NOT NULL,  
    last_name VARCHAR(50) NOT NULL,  
    email VARCHAR(100) UNIQUE NOT NULL,  
    password VARCHAR(255) NOT NULL,  
    phone VARCHAR(20),  
    address TEXT,  
    created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP,  
    INDEX idx_email (email)  
);
```

```
-- Table 3: Product
```

```
CREATE TABLE Product (  
    product_id INT PRIMARY KEY AUTO_INCREMENT,  
    product_name VARCHAR(100) NOT NULL,  
    category_id INT,  
    price DECIMAL(10,2) NOT NULL CHECK (price >= 0),  
    stock_quantity INT NOT NULL DEFAULT 0 CHECK (stock_quantity >= 0),  
    FOREIGN KEY (category_id) REFERENCES Category(category_id),  
    INDEX idx_product_name (product_name)  
);
```

-- Table 4: User (for admin/staff)

```
CREATE TABLE User (  
    user_id INT PRIMARY KEY AUTO_INCREMENT,  
    username VARCHAR(50) UNIQUE NOT NULL,  
    password VARCHAR(255) NOT NULL,  
    role ENUM('admin', 'staff', 'viewer') DEFAULT 'staff'  
);
```

-- Table 5: Order

```
CREATE TABLE `Order` (  
    order_id INT PRIMARY KEY AUTO_INCREMENT,  
    customer_id INT NOT NULL,  
    order_date TIMESTAMP DEFAULT CURRENT_TIMESTAMP,  
    shipping_address TEXT NOT NULL,  
    total_amount DECIMAL(10,2) DEFAULT 0,  
    status ENUM('pending', 'processing', 'shipped', 'delivered', 'cancelled') DEFAULT  
'pending',  
    FOREIGN KEY (customer_id) REFERENCES Customer(customer_id),  
    INDEX idx_customer_id (customer_id),  
    INDEX idx_order_date (order_date)  
);
```

-- Table 6: OrderItem

```
CREATE TABLE OrderItem (  
    order_item_id INT PRIMARY KEY AUTO_INCREMENT,  
    order_id INT NOT NULL,  
    product_id INT NOT NULL,  
    quantity INT NOT NULL CHECK (quantity > 0),  
    unit_price DECIMAL(10,2) NOT NULL,  
    subtotal DECIMAL(10,2) GENERATED ALWAYS AS (quantity * unit_price) STORED,  
    FOREIGN KEY (order_id) REFERENCES `Order`(order_id) ON DELETE CASCADE,  
    FOREIGN KEY (product_id) REFERENCES Product(product_id),  
    UNIQUE KEY unique_order_product (order_id, product_id)  
);
```

-- Table 7: Payment

```
CREATE TABLE Payment (  
    payment_id INT PRIMARY KEY AUTO_INCREMENT,  
    order_id INT NOT NULL,  
    payment_date TIMESTAMP DEFAULT CURRENT_TIMESTAMP,  
    amount DECIMAL(10,2) NOT NULL CHECK (amount > 0),
```

```

    payment_method ENUM('Credit Card', 'GCash', 'PayMaya', 'COD') NOT NULL,
    payment_status ENUM('Pending', 'Completed', 'Failed', 'Refunded') DEFAULT
'Pending',
    transaction_reference VARCHAR(100) UNIQUE,
    FOREIGN KEY (order_id) REFERENCES `Order`(order_id),
    INDEX idx_order_id (order_id)
);

```

-- Table 8: AuditLog

```

CREATE TABLE AuditLog (
    log_id INT PRIMARY KEY AUTO_INCREMENT,
    table_name VARCHAR(50),
    action VARCHAR(20),
    record_id INT,
    old_data TEXT,
    new_data TEXT,
    changed_by VARCHAR(50),
    change_date TIMESTAMP DEFAULT CURRENT_TIMESTAMP
);

```

-- Initial Data

```

INSERT INTO Category (category_name) VALUES ('Electronics'), ('Clothing'),
('Books');

```

```

INSERT INTO User (username, password, role) VALUES
('admin', MD5('admin123'), 'admin');

```

```

SHOW TABLES;

```

Database Schema and Table Descriptions

Category Table

Column	Type	Constraints	Description
category_id	INT	PRIMARY KEY, AUTO_INCREMENT	Unique category identifier
category_name	VARCHAR(50)	UNIQUE, NOT NULL	Name of product category

description	TEXT	-	Category details
-------------	------	---	------------------

Customer Table

Column	Type	Constraints	Description
customer_id	INT	PRIMARY KEY, AUTO_INCREMENT	Unique customer identifier
first_name	VARCHAR(50)	NOT NULL	Customer's first name
last_name	VARCHAR(50)	NOT NULL	Customer's last name
email	VARCHAR(100)	UNIQUE, NOT NULL	Customer's email address
password	VARCHAR(255)	NOT NULL	Hashed password
phone	VARCHAR(20)	-	Contact number
address	TEXT	-	Shipping address
created_at	TIMESTAMP	DEFAULT CURRENT_TIMESTAMP	Account creation date

Product Table

Column	Type	Constraints	Description
product_id	INT	PRIMARY KEY, AUTO_INCREMENT	Unique product identifier
product_name	VARCHAR(100)	NOT NULL	Name of product
category_id	INT	FOREIGN KEY → Category(category_id)	Product category
price	DECIMAL(10,2)	NOT NULL, CHECK (price >= 0)	Product price
stock_quantity	INT	NOT NULL, DEFAULT 0, CHECK (stock_quantity >= 0)	Available stock

		>= 0)	
--	--	-------	--

User Table

Column	Type	Constraints	Description
user_id	INT	PRIMARY KEY, AUTO_INCREMENT	Unique user identifier
username	VARCHAR(50)	UNIQUE, NOT NULL	Login username
password	VARCHAR(255)	NOT NULL	Hashed password
role	ENUM	DEFAULT 'staff'	admin/staff/viewer

Order Table

Column	Type	Constraints	Description
order_id	INT	PRIMARY KEY, AUTO_INCREMENT	Unique order identifier
customer_id	INT	NOT NULL, FOREIGN KEY → Customer(customer_id)	Customer who placed order
order_date	TIMESTAMP	DEFAULT CURRENT_TIMESTAMP	Date order was placed
shipping_address	TEXT	NOT NULL	Delivery address
total_amount	DECIMAL(10,2)	DEFAULT 0	Order total amount
status	ENUM	DEFAULT 'pending'	pending/processing/shipped/delivered/cancelled

OrderItem Table

Column	Type	Constraints	Description
order_item_id	INT	PRIMARY KEY, AUTO_INCREMENT	Unique item identifier
order_id	INT	NOT NULL, FOREIGN KEY → Order(order_id), ON DELETE CASCADE	Associated order
product_id	INT	NOT NULL, FOREIGN KEY → Product(product_id)	Product ordered
quantity	INT	NOT NULL, CHECK (quantity > 0)	Number of units
unit_price	DECIMAL(10,2)	NOT NULL	Price at time of order
subtotal	DECIMAL(10,2)	GENERATED ALWAYS AS (quantity * unit_price) STORED	Calculated total for item

Payment Table

Column	Type	Constraints	Description
payment_id	INT	PRIMARY KEY, AUTO_INCREMENT	Unique payment identifier
order_id	INT	NOT NULL, FOREIGN KEY → Order(order_id)	Associated order
payment_date	TIMESTAMP	DEFAULT	When payment was

		CURRENT_TIMES TAMP	made
amount	DECIMAL(10, 2)	NOT NULL, CHECK (amount > 0)	Payment amount
payment_method	ENUM	NOT NULL	Credit Card/GCash/PayMa ya/COD
payment_status	ENUM	DEFAULT 'Pending'	Pending/Completed /Failed/Refunded
transaction_reference	VARCHAR(10 0)	UNIQUE	Reference number for transaction

AuditLog Table

Column	Type	Constraints	Description
log_id	INT	PRIMARY KEY, AUTO_INCREMENT	Unique log identifier
table_name	VARCHAR(50)	–	Table that was modified
action	VARCHAR(20)	-	INSERT/UPDATE/DE LETE
record_id	INT	-	ID of affected record
old_data	TEXT	-	Previous values before change
new_data	TEXT	-	New values after change
changed_by	VARCHAR(50)	-	User who made the change
change_date	TIMESTAMP	DEFAULT CURRENT_TIMESTA MP	When change occurred



Sample SQL queries (basic, BI, optimized)

BASIC QUERIES

-- Total sales per customer

```
SELECT c.customer_id, c.first_name, c.last_name, SUM(o.total_amount) as total_spent
FROM Customer c
JOIN `Order` o ON c.customer_id = o.customer_id
GROUP BY c.customer_id, c.first_name, c.last_name
ORDER BY total_spent DESC
LIMIT 20;
```

-- Monthly sales report

```
SELECT DATE_FORMAT(order_date, '%Y-%m') as month,
COUNT(*) as total_orders,
SUM(total_amount) as revenue
FROM `Order`
GROUP BY DATE_FORMAT(order_date, '%Y-%m')
ORDER BY month DESC;
```

-- Top selling products

```
SELECT p.product_id, p.name,
SUM(oi.quantity) as total_sold,
SUM(oi.subtotal) as total_revenue
FROM Product p
JOIN OrderItem oi ON p.product_id = oi.product_id
GROUP BY p.product_id, p.name
ORDER BY total_sold DESC
LIMIT 10;
```

-- Customer order history

```
SELECT c.customer_id, c.first_name, c.last_name,
COUNT(o.order_id) as order_count,
SUM(o.total_amount) as total_spent
```



```
HAVING order_count > 0  
ORDER BY total_spent DESC;
```

```
-- Pending orders  
SELECT o.order_id, o.order_date, o.total_amount,  
c.first_name, c.last_name, c.email  
FROM `Order` o  
JOIN Customer c ON o.customer_id = c.customer_id  
WHERE o.status = 'pending'  
ORDER BY o.order_date DESC;
```

BI QUERIES

```
-- Total sales per customer (BI view)  
SELECT c.customer_id, c.first_name, c.last_name,  
SUM(o.total_amount) as total_spent  
FROM Customer c  
INNER JOIN `Order` o ON c.customer_id = o.customer_id  
GROUP BY c.customer_id  
ORDER BY total_spent DESC  
LIMIT 20;
```

```
-- Monthly revenue report  
SELECT DATE_FORMAT(order_date, '%Y-%m') as month,  
COUNT(*) as total_orders,  
SUM(total_amount) as revenue  
FROM `Order`  
GROUP BY month  
ORDER BY month DESC;
```

```
-- Top products performance  
SELECT p.product_id, p.name,  
SUM(oi.quantity) as total_sold,  
SUM(oi.quantity * oi.unit_price) as total_revenue  
FROM Product p
```



LIMIT 10;

OPTIMIZED QUERIES

-- Optimized customer sales query

```
SELECT c.customer_id, c.first_name, c.last_name,  
SUM(o.total_amount) as total_spent  
FROM Customer c  
INNER JOIN `Order` o ON c.customer_id = o.customer_id  
GROUP BY c.customer_id  
ORDER BY total_spent DESC  
LIMIT 20;
```

-- Optimized monthly sales report

```
SELECT DATE_FORMAT(order_date, '%Y-%m') as month,  
COUNT(*) as total_orders,  
SUM(total_amount) as revenue  
FROM `Order`  
WHERE order_date IS NOT NULL  
GROUP BY month  
ORDER BY month DESC;
```

-- Optimized product sales query

```
SELECT p.product_id, p.name,  
SUM(oi.quantity) as total_sold,  
SUM(oi.quantity * oi.unit_price) as total_revenue  
FROM Product p  
INNER JOIN OrderItem oi ON p.product_id = oi.product_id  
GROUP BY p.product_id, p.name  
ORDER BY total_sold DESC  
LIMIT 10;
```

-- Optimized pending orders

```
SELECT o.order_id, o.order_date, o.total_amount,  
c.first_name, c.last_name, c.email  
FROM `Order` o
```



BACKUP AND RECOVERY STEPS

```
-- MariaDB dump 10.19 Distrib 10.4.32-MariaDB, for Win64 (AMD64)
```

```
--
```

```
-- Host: localhost Database: online_shop
```

```
-----
```

```
-- Server version 10.4.32-MariaDB
```

```
/*!40101 SET @OLD_CHARACTER_SET_CLIENT=@@CHARACTER_SET_CLIENT */;
```

```
/*!40101 SET @OLD_CHARACTER_SET_RESULTS=@@CHARACTER_SET_RESULTS */;
```

```
/*!40101 SET @OLD_COLLATION_CONNECTION=@@COLLATION_CONNECTION */;
```

```
/*!40101 SET NAMES utf8mb4 */;
```

```
/*!40103 SET @OLD_TIME_ZONE=@@TIME_ZONE */;
```

```
/*!40103 SET TIME_ZONE='+00:00' */;
```

```
/*!40014 SET @OLD_UNIQUE_CHECKS=@@UNIQUE_CHECKS, UNIQUE_CHECKS=0 */;
```

```
/*!40014 SET @OLD_FOREIGN_KEY_CHECKS=@@FOREIGN_KEY_CHECKS, FOREIGN_KEY_CHECKS=0 */;
```

```
/*!40101 SET @OLD_SQL_MODE=@@SQL_MODE, SQL_MODE='NO_AUTO_VALUE_ON_ZERO' */;
```

```
/*!40111 SET @OLD_SQL_NOTES=@@SQL_NOTES, SQL_NOTES=0 */;
```

```
--
```

```
-- Table structure for table `auditlog`
```

```
--
```

```
DROP TABLE IF EXISTS `auditlog`;
```

```
/*!40101 SET @saved_cs_client = @@character_set_client */;
```

```
/*!40101 SET character_set_client = utf8 */;
```

```
CREATE TABLE `auditlog` (
```

```
  `log_id` int(11) NOT NULL AUTO_INCREMENT,
```

```
  `table_name` varchar(50) DEFAULT NULL,
```

```
  `action` varchar(20) DEFAULT NULL,
```

```
  `record_id` int(11) DEFAULT NULL,
```

```
  `old_data` text DEFAULT NULL,
```

```
  `new_data` text DEFAULT NULL,
```



```
/*!40101 SET character_set_client = @saved_cs_client */;

--
-- Dumping data for table `auditlog`
--

LOCK TABLES `auditlog` WRITE;
/*!40000 ALTER TABLE `auditlog` DISABLE KEYS */;
/*!40000 ALTER TABLE `auditlog` ENABLE KEYS */;
UNLOCK TABLES;

--
-- Table structure for table `customer`
--
DROP TABLE IF EXISTS `customer`;
/*!40101 SET @saved_cs_client = @@character_set_client */;
/*!40101 SET character_set_client = utf8 */;
CREATE TABLE `customer` (
  `customer_id` int(11) NOT NULL AUTO_INCREMENT,
  `first_name` varchar(50) NOT NULL,
  `last_name` varchar(50) NOT NULL,
  `email` varchar(100) NOT NULL,
  `password` varchar(255) NOT NULL,
  `phone` varchar(20) DEFAULT NULL,
  `address` text DEFAULT NULL,
  `created_at` timestamp NOT NULL DEFAULT current_timestamp(),
  PRIMARY KEY (`customer_id`),
  UNIQUE KEY `email` (`email`),
  KEY `idx_email` (`email`)
) ENGINE=InnoDB DEFAULT CHARSET=utf8mb4 COLLATE=utf8mb4_general_ci;
/*!40101 SET character_set_client = @saved_cs_client */;

--
-- Dumping data for table `customer`
--

LOCK TABLES `customer` WRITE;
/*!40000 ALTER TABLE `customer` DISABLE KEYS */;
```



-- Table structure for table `order`

--

```
DROP TABLE IF EXISTS `order`;  
/*!40101 SET @saved_cs_client = @@character_set_client */;  
/*!40101 SET character_set_client = utf8 */;  
CREATE TABLE `order` (  
  `order_id` int(11) NOT NULL AUTO_INCREMENT,  
  `customer_id` int(11) NOT NULL,  
  `order_date` timestamp NOT NULL DEFAULT current_timestamp(),  
  `total_amount` decimal(10,2) NOT NULL DEFAULT 0.00,  
  `status` enum('pending','processing','shipped','delivered','cancelled') DEFAULT 'pending',  
  PRIMARY KEY (`order_id`),  
  KEY `idx_customer_id` (`customer_id`),  
  KEY `idx_order_date` (`order_date`),  
  CONSTRAINT `order_ibfk_1` FOREIGN KEY (`customer_id`) REFERENCES `customer` (`customer_id`)  
) ENGINE=InnoDB DEFAULT CHARSET=utf8mb4 COLLATE=utf8mb4_general_ci;  
/*!40101 SET character_set_client = @saved_cs_client */;
```

--

-- Dumping data for table `order`

--

```
LOCK TABLES `order` WRITE;  
/*!40000 ALTER TABLE `order` DISABLE KEYS */;  
/*!40000 ALTER TABLE `order` ENABLE KEYS */;  
UNLOCK TABLES;
```

--

-- Table structure for table `orderitem`

--

```
DROP TABLE IF EXISTS `orderitem`;  
/*!40101 SET @saved_cs_client = @@character_set_client */;  
/*!40101 SET character_set_client = utf8 */;  
CREATE TABLE `orderitem` (  
  `order_item_id` int(11) NOT NULL AUTO_INCREMENT,
```



```
unit_price` decimal(10,2) NOT NULL,  
`subtotal` decimal(10,2) GENERATED ALWAYS AS (`quantity` * `unit_price`) STORED,  
  
PRIMARY KEY (`order_item_id`),  
UNIQUE KEY `unique_order_product` (`order_id`,`product_id`),  
KEY `idx_order_id` (`order_id`),  
KEY `idx_product_id` (`product_id`),  
CONSTRAINT `orderitem_ibfk_1` FOREIGN KEY (`order_id`) REFERENCES `order` (`order_id`) ON DELETE  
CASCADE,  
CONSTRAINT `orderitem_ibfk_2` FOREIGN KEY (`product_id`) REFERENCES `product` (`product_id`)  
) ENGINE=InnoDB DEFAULT CHARSET=utf8mb4 COLLATE=utf8mb4_general_ci;  
/*!40101 SET character_set_client = @saved_cs_client */;  
  
--  
-- Dumping data for table `orderitem`  
--  
LOCK TABLES `orderitem` WRITE;  
/*!40000 ALTER TABLE `orderitem` DISABLE KEYS */;  
/*!40000 ALTER TABLE `orderitem` ENABLE KEYS */;  
UNLOCK TABLES;  
  
--  
-- Table structure for table `product`  
--  
DROP TABLE IF EXISTS `product`;  
/*!40101 SET @saved_cs_client = @@character_set_client */;  
/*!40101 SET character_set_client = utf8 */;  
CREATE TABLE `product` (  
  `product_id` int(11) NOT NULL AUTO_INCREMENT,  
  `name` varchar(100) NOT NULL,  
  `description` text DEFAULT NULL,  
  `price` decimal(10,2) NOT NULL CHECK (`price` >= 0),  
  `stock_quantity` int(11) NOT NULL DEFAULT 0 CHECK (`stock_quantity` >= 0),  
  `category` varchar(50) DEFAULT NULL,  
  `created_at` timestamp NOT NULL DEFAULT current_timestamp(),  
  PRIMARY KEY (`product_id`),  
  KEY `idx_category` (`category`),  
  KEY `idx_price` (`price`)  
) ENGINE=InnoDB DEFAULT CHARSET=utf8mb4 COLLATE=utf8mb4_general_ci;
```



```
/*!40101 SET character_set_client = @saved_cs_client */;

--
-- Dumping data for table `product`
--
LOCK TABLES `product` WRITE;
/*!40000 ALTER TABLE `product` DISABLE KEYS */;
/*!40000 ALTER TABLE `product` ENABLE KEYS */;
UNLOCK TABLES;

--
-- Table structure for table `user`
--
DROP TABLE IF EXISTS `user`;
/*!40101 SET @saved_cs_client = @@character_set_client */;
/*!40101 SET character_set_client = utf8 */;
CREATE TABLE `user` (
  `user_id` int(11) NOT NULL AUTO_INCREMENT,
  `username` varchar(50) NOT NULL,
  `password` varchar(255) NOT NULL,
  `role` enum('admin','staff','viewer') DEFAULT 'staff',
  `created_at` timestamp NOT NULL DEFAULT current_timestamp(),
  PRIMARY KEY (`user_id`),
  UNIQUE KEY `username` (`username`),
  KEY `idx_username` (`username`)
) ENGINE=InnoDB AUTO_INCREMENT=4 DEFAULT CHARSET=utf8mb4 COLLATE=utf8mb4_general_ci;
/*!40101 SET character_set_client = @saved_cs_client */;

--
-- Dumping data for table `user`
--
LOCK TABLES `user` WRITE;
/*!40000 ALTER TABLE `user` DISABLE KEYS */;
INSERT INTO `user` VALUES (1,'admin','0192023a7bbd73250516f069df18b500','admin','2026-04-16
15:22:18'),(2,'staff1','de9bf5643eabf80f4a56fda3bbb84483','staff','2026-04-16
15:22:18'),(3,'viewer1','49e5e739ea41d635246cd9cd21af17c4','viewer','2026-04-16 15:22:18');
/*!40000 ALTER TABLE `user` ENABLE KEYS */;
UNLOCK TABLES;
/*!40103 SET TIME_ZONE=@OLD_TIME_ZONE */;
/*!40101 SET SQL_MODE=@OLD_SQL_MODE */;
```




```
/*!40014 SET UNIQUE_CHECKS=@OLD_UNIQUE_CHECKS */;/*!40101 SET  
CHARACTER_SET_CLIENT=@OLD_CHARACTER_SET_CLIENT */;  
/*!40101 SET CHARACTER_SET_RESULTS=@OLD_CHARACTER_SET_RESULTS */;  
/*!40101 SET COLLATION_CONNECTION=@OLD_COLLATION_CONNECTION */;  
/*!40111 SET SQL_NOTES=@OLD_SQL_NOTES */;
```

-- Dump completed on 2026-04-16 23:34:43

RESTORE PROOF

```
C:\Program Files\MySQL\MySQL Server 9.6\bin>mysql -u root -e "USE onlineShop; SHOW TABLES;"  
+-----+  
| Tables_in_onlineshop |  
+-----+  
| auditlog  
| category  
| customer  
| order  
| orderitem  
| payment  
| product  
| user  
+-----+
```



Indexing and performance improvements

Before:

```
MariaDB [online_shop]> USE online_shop;
Database changed
MariaDB [online_shop]> CREATE INDEX idx_order_customer ON `Order`(customer_id);
Query OK, 0 rows affected, 1 warning (0.144 sec)
Records: 0 Duplicates: 0 Warnings: 1

MariaDB [online_shop]> CREATE INDEX idx_customer_id ON Customer(customer_id);CREATE INDEX idx_order_date ON `Order`(order_date);
Query OK, 0 rows affected (0.235 sec)
Records: 0 Duplicates: 0 Warnings: 0

ERROR 1061 (42000): Duplicate key name 'idx_order_date'
MariaDB [online_shop]> CREATE INDEX idx_order_status ON `Order`(status);CREATE INDEX idx_orderitem_product ON OrderItem(product_id);
Query OK, 0 rows affected (0.124 sec)
Records: 0 Duplicates: 0 Warnings: 0

Query OK, 0 rows affected, 1 warning (0.307 sec)
Records: 0 Duplicates: 0 Warnings: 1

MariaDB [online_shop]> CREATE INDEX idx_orderitem_order ON OrderItem(order_id);
Query OK, 0 rows affected, 1 warning (0.249 sec)
Records: 0 Duplicates: 0 Warnings: 1

MariaDB [online_shop]> CREATE INDEX idx_product_id ON Product(product_id);CREATE INDEX idx_order_status_date ON `Order`(status, order_date);
Query OK, 0 rows affected (0.022 sec)
Records: 0 Duplicates: 0 Warnings: 0

Query OK, 0 rows affected (0.135 sec)
Records: 0 Duplicates: 0 Warnings: 0
```

After:

```
MariaDB [online_shop]> SHOW INDEX FROM Customer;
```

Table	Non_unique	Key_name	Seq_in_index	Column_name	Collation	Cardinality	Sub_part	Packed	Null	Index_type	Comment	Index_comment
customer	0	PRIMARY	1	customer_id	A	99085	NULL	NULL		BTREE		
customer	0	email	1	email	A	99085	NULL	NULL		BTREE		
customer	1	idx_email	1	email	A	99085	NULL	NULL		BTREE		
customer	1	idx_customer_id	1	customer_id	A	99085	NULL	NULL		BTREE		

4 rows in set (0.004 sec)

```
MariaDB [online_shop]> SHOW INDEX FROM `Order`;
```

Table	Non_unique	Key_name	Seq_in_index	Column_name	Collation	Cardinality	Sub_part	Packed	Null	Index_type	Comment	Index_comment
order	0	PRIMARY	1	order_id	A	48961	NULL	NULL		BTREE		
order	1	idx_customer_id	1	customer_id	A	48961	NULL	NULL		BTREE		
order	1	idx_order_date	1	order_date	A	730	NULL	NULL		BTREE		
order	1	idx_order_customer	1	customer_id	A	48961	NULL	NULL		BTREE		
order	1	idx_order_status	1	status	A	10	NULL	NULL	YES	BTREE		
order	1	idx_order_status_date	1	status	A	10	NULL	NULL	YES	BTREE		
order	1	idx_order_status_date	2	order_date	A	3766	NULL	NULL		BTREE		

7 rows in set (0.001 sec)

```
MariaDB [online_shop]> SHOW INDEX FROM Product;
```

Table	Non_unique	Key_name	Seq_in_index	Column_name	Collation	Cardinality	Sub_part	Packed	Null	Index_type	Comment	Index_comment
orderitem	0	PRIMARY	1	order_item_id	A	149184	NULL	NULL		BTREE		
orderitem	0	unique_order_product	1	order_id	A	149184	NULL	NULL		BTREE		
orderitem	0	unique_order_product	2	product_id	A	149184	NULL	NULL		BTREE		
orderitem	1	idx_order_id	1	order_id	A	149184	NULL	NULL		BTREE		
orderitem	1	idx_product_id	1	product_id	A	18	NULL	NULL		BTREE		
orderitem	1	idx_orderitem_product	1	product_id	A	18	NULL	NULL		BTREE		
orderitem	1	idx_orderitem_order	1	order_id	A	149184	NULL	NULL		BTREE		

7 rows in set (0.001 sec)



Screenshots of outputs and dashboards

- Sample_Query_Result

Showing rows 0 - 9 (10 total, Query took 0.0194 seconds.)

```
SELECT o.order_id, c.first_name, c.last_name, o.order_date, o.total_amount, o.status FROM `Order` o JOIN Customer c ON o.customer_id = c.customer_id LIMIT 10;
```

☐ Profiling [Edit inline] [Edit] [Explain SQL] [Create PHP code] [Refresh]

Extra options

order_id	first_name	last_name	order_date	total_amount	status
1	Thomas	Allen	2026-02-02 23:44:12	0.00	pending
2	Brian	Wilson	2025-12-20 23:44:12	0.00	pending
4	Betty	Thomas	2026-01-01 23:45:18	0.00	pending
6	Andrew	Moore	2025-12-09 23:46:01	0.00	pending
7	First40121	Last40121	2026-01-28 23:46:01	0.00	pending
8	David	Parker	2026-01-11 23:46:01	0.00	pending
10	Elizabeth	Anderson	2025-11-09 23:47:57	0.00	pending
11	Sarah	King	2025-11-06 23:47:57	0.00	pending
12	Anthony	Phillips	2026-02-14 23:47:57	0.00	shipped
13	Sandra	Smith	2026-01-11 23:47:57	0.00	pending

Query results operations

- Optimized

```
MariaDB [online_shop]> SELECT c.customer_id, c.first_name, c.last_name, SUM(o.total_amount) as total_spent
-> FROM Customer c
-> INNER JOIN `Order` o ON c.customer_id = o.customer_id
-> GROUP BY c.customer_id
-> ORDER BY total_spent DESC
-> LIMIT 20;
```

customer_id	first_name	last_name	total_spent
19159	First19159	Last19159	510.00
37763	First37763	Last37763	509.99
1404	First1404	Last1404	509.99
54744	First54744	Last54744	509.96
44879	First44879	Last44879	509.95
28818	First28818	Last28818	509.92
18194	First18194	Last18194	509.90
36220	First36220	Last36220	509.90
1260	First1260	Last1260	509.88
37874	First37874	Last37874	509.87
15507	First15507	Last15507	509.87
46040	First46040	Last46040	509.86
24288	First24288	Last24288	509.86
11202	First11202	Last11202	509.84
28507	First28507	Last28507	509.84
3575	First3575	Last3575	509.83
31030	First31030	Last31030	509.83
32977	First32977	Last32977	509.83
53276	First53276	Last53276	509.82
49396	First49396	Last49396	509.81

20 rows in set (0.400 sec)



```
MariaDB [online_shop]> SELECT DATE_FORMAT(order_date, '%Y-%m') as month,
->      COUNT(*) as total_orders,
->      SUM(total_amount) as revenue
-> FROM `Order`
-> WHERE order_date IS NOT NULL
-> GROUP BY DATE_FORMAT(order_date, '%Y-%m')
-> ORDER BY month DESC;
```

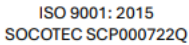
month	total_orders	revenue
2026-05	244	60991.19
2026-04	4108	1061340.69
2026-03	4270	1114990.29
2026-02	3846	1004318.23
2026-01	4292	1107395.94
2025-12	4210	1090478.11
2025-11	4211	1097636.46
2025-10	4221	1092704.88
2025-09	4100	1075886.69
2025-08	4156	1074668.59
2025-07	4150	1084304.30
2025-06	4094	1068515.17
2025-05	4098	1067731.87

13 rows in set (0.111 sec)

```
MariaDB [online_shop]> SELECT p.product_id, p.name,
->      SUM(oi.quantity) as total_sold,
->      SUM(oi.quantity * oi.unit_price) as total_revenue
-> FROM Product p
-> INNER JOIN OrderItem oi ON p.product_id = oi.product_id
-> GROUP BY p.product_id, p.name
-> ORDER BY total_sold DESC
-> LIMIT 10;
```

product_id	name	total_sold	total_revenue
4	USB-C Cable	45216	587355.84
9	Noise Cancelling Headphones	45043	9008149.57
8	External SSD	44806	6720451.94
7	Laptop Stand	44803	1791671.97
3	Mechanical Keyboard	44650	5804053.50
5	Monitor 27"	44645	15625303.55
2	Wireless Mouse	44618	1338093.82
6	Webcam HD	44489	2668895.11
1	Laptop Pro	44447	40001855.53
10	Gaming Chair	44230	13268557.70

10 rows in set (3.847 sec)



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```
10120 rows in set (0.269 sec)
```



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Reflection and limitations



Role Assignments

- Abby Llaguno – Database Designer
- Donalen Grace Rico – SQL Developer
- Eunice Lozano – Security Officer
- Sandy Murillo – Project Manager
- Annaliza Juarez – Documentation Lead
- Gabe Arcenal – QA/Tester

Contribution Summary

Abby Llaguno (Database Designer)

- Designed the initial and final ER diagram
- Defined entities, attributes, and relationships
- Ensured proper normalization of the database structure
- Created the database schema and table structure

Donalen Grace Rico (SQL Developer)

- Developed basic SQL queries (SELECT, JOIN, GROUP BY)
- Created Business Intelligence (BI) queries for reporting
- Optimized queries for better performance
- Assisted in testing query outputs

Eunice Lozano (Security Officer)

- Designed backup and recovery procedures
- Implemented data security measures
- Managed user roles and access control
- Ensured data integrity and protection
- Ensured data integrity and protection

Sandy Murillo (Project Manager)

- Led project planning and task distribution
- Monitored group progress and deadlines
- Coordinated communication among members
- Ensured completion of all required outputs



Annaliza Juarez (Documentation Lead)

- Compiled all project documentation
- Organized the final PDF document
- Wrote descriptions for database schema and queries
- Added screenshots and formatted the document

Gabe Arcenal (QA/Tester)

- Tested database functionality and queries
- Identified and reported errors or issues
- Verified system performance with large datasets
- Ensured accuracy of outputs and reports



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